Ryan Nepal

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EDUCATION

University of Massachusetts Amherst Expected May 2028
BS in Mechanical Engineering
Coursework: Machine Design, Thermodynamics, Fluid Mechanics, Dynamics, Manufacturing Processes, Materials Science

EXPERIENCE

Amazon | Reliability and Maintenance Engineering Intern June 2026 - August 2026
Las Vegas, NV

- Designed and validated a PLC-controlled tote-turner using pneumatic actuators and photoelectric sensors to automate tote orientation on conveyance; leveraged industrial automation, ladder logic, and GD&T — increased accuracy **14%**
- Built and deployed a full-stack web application on AWS (Elastic Beanstalk, PostgreSQL, S3) with Claude Code, for tracking conveyor sorter amnesty across a site; first-of-its kind system enabling data-driven root cause analysis.
- Owned DMAIC project from baseline through launch; **700+ units** logged in first **10 days** with projected **2-4 hours/week** in time savings through elimination of ad hoc investigation and jam prevention.

NexTek, LLC | Mechanical Engineering Intern May 2025 - August 2025
Billerica, MA

- Owned design, assembly, and validation of a **16-up** leak test fixture to withstand **680 lbf** for IP-rated RF surge protectors; performed FEA to evaluate assembly strength; tested **21,000+ units** and increased efficiency by **300%**.
- Supported end-to-end design and validation of EMI/RF protection components; applied CAD/3D modeling, GD&T, assembly, and bench testing to verify performance and reliability.
- Managed **~4** concurrent projects; led troubleshooting/failure analysis and drove ECOs/change control to resolve build issues and maintain spec compliance.

PROJECTS

Rebuild of a Salvage 2008 Mazda3 S January 2026

- Performed a comprehensive mechanical and structural assessment, identifying front-end collision damage and engine performance issues through component inspection and fault-code diagnostics.
- Reconstructed the front-end assembly and engineered a repair for sheared intake manifold ports by precision tapping and installing vacuum valves and lines with high-temp epoxy; restored proper airflow and eliminated engine misfires.

Design, Simulation, and CNC Machining of a Fuel Pump Adapter November 2025

- Designed a custom aluminum fuel pump adapter to mechanically interface an NPT hose connector with a pressurized fuel reservoir and fuel pump; used ANSYS to validate structural integrity through pressure and stress analysis.
- Programmed and CNC-machined the adapter from stock material using Autodesk Fusion; performed deburring, fitment verification, and pressure testing to validate leak-free operation.

LEADERSHIP

UMass Supermileage Vehicle Team | Fuel System Lead September 2025 - Present

- Owned design and optimization of fuel delivery system; utilized CAD and FEA to improve combustion efficiency and structural integrity; applied CNC machining to develop efficient components; achieved **414 mpg** in competition.
- Collaborate with a multidisciplinary engineering team to design, develop, and optimize a high-efficiency vehicle for competition at the Shell Eco-marathon at the Indianapolis Motor Speedway.

UMass Transit Services | CDL Instructor, Dispatcher, Operator September 2024 - Present

- Train Class B CDL-permit drivers in safe operation, defensive driving, and transit procedures while instructing on bus mechanical systems to ensure thorough understanding of the vehicle.
- Direct day-to-day operations of a 28-bus fleet, supervising drivers, coordinating dispatch, and resolving service disruptions to ensure safe, reliable, and on-time transit service.

SKILLS & TOOLS

Software: SolidWorks, ANSYS, MATLAB, Inventor, Fusion, AutoCAD, PLC Programming, RSLogix 5000, MS Excel
Engineering: CAD / 3D Modeling, DFMA, FEA, CNC Machining, GD&T, Root Cause Analysis, DMAIC, Prototyping